



Computer Engineering Dept.
Faculty of Engineering
Cairo University



Machine Learning Assignment1

Rawan Mostafa Mahmoud

Sec1

BN:18

ID:9210423

Machine Learning: Assignment 1

Problem 1.6 :

$$N=10 \quad p(\text{red}) = \mu$$

(a) 1 Sample

for one marble: $p(\text{not red}) = 1 - \mu$

$\because$ each marble is independent on the others

$$\therefore p(v=0) = (1 - \mu)^N$$

@ $\mu = 0.05$:

$$p(v=0) = (1 - 0.05)^{10} = \boxed{0.5987}$$

@ $\mu = 0.5$:

$$p(v=0) = (1 - 0.5)^{10} = \boxed{9.7656 \times 10^{-4}}$$

@ $\mu = 0.8$:

$$p(v=0) = (1 - 0.8)^{10} = \boxed{1.024 \times 10^{-7}}$$

(b) 1000 Samples, req: $p(\text{at least 1 sample has } v=0)$

$p(\text{at least 1 sample has } v=0)$

$$= p(1, 2, 3, \dots, N \text{ has } v=0)$$

$$= 1 - p(\text{No sample has } v=0)$$

$$= 1 - p(\text{all samples have } v \neq 0)$$

$$= 1 - [1 - p(v=0)]^{1000} \leftarrow$$

@ $\mu = 0.05$:

$$p = 1 - [1 - 0.5987]^{1000} = \boxed{1}$$

@ $\mu = 0.5$:

$$p = 1 - [1 - 9.7656 \times 10^{-4}]^{1000} = \boxed{0.6236}$$

@ $\mu = 0.8$:

$$p = 1 - [1 - 1.024 \times 10^{-7}]^{1000} = \boxed{1.0239 \times 10^{-4}}$$

(c) For 1000 000 Independent samples:

@ $\alpha = 0.05$:

$$p = 1 - (1 - 0.5987)^{1000\ 000} = \boxed{1}$$

@ $\alpha = 0.5$:

$$p = 1 - (1 - 9.7656 \times 10^{-4})^{1000\ 000} = \boxed{1}$$

@ $\alpha = 0.8$:

$$p = 1 - (1 - 1.024 \times 10^{-7})^{1000\ 000} = \boxed{0.09733}$$

Problem (2.5):

Prove By Induction that: $\sum_{i=0}^D \binom{N}{i} \leq N+1$

, hence $m_H(N) \leq N^{dvc} + 1$

- Base Case:

@ $D=0$

$$N C_0 \leq N^0 + 1$$

$1 \leq 2$ ✓ true (for any N)

@ $N=1$

$$1 C_0 \leq 1^0 + 1$$

& $D=0$

$1 \leq 2$ ✓ true

* Assume statement is true for all $N \leq N_0$ for any D

* Req. to prove: for $N = N_0 + 1$, the statement is true. $\leftarrow$

$$\hookrightarrow \sum_{i=0}^D \binom{N_0+1}{i} \leq (N_0+1)^D + 1 \quad \leftarrow \text{Req. to prove.}$$

Knowing that: $\binom{N_0+1}{i} = \binom{N_0}{i} + \binom{N_0}{i-1}$

$$\begin{aligned} \therefore \sum_{i=0}^D \binom{N_0+1}{i} &= \sum_{i=0}^D \binom{N_0}{i} + \sum_{i=0}^D \binom{N_0}{i-1} \\ &= \sum_{i=0}^D \binom{N_0}{i} + \sum_{i=0}^{D-1} \binom{N_0}{i} \end{aligned}$$

$$\leq N_0^D + 1 + N_0^{D-1} + 1$$

$$\leq (N_0^D + N_0^{D-1} + 1) + 1 \leq (N_0+1)^D + 1 \quad \#$$

$$\therefore \sum_{i=0}^D \binom{N}{i} \leq N^D + 1 \quad \#$$

$$\text{and } \therefore m_H(N) \leq \sum_{i=0}^{dvc} \binom{N}{i} \leq N^{dvc} + 1$$

$$\therefore m_H(N) \leq N^{dvc} + 1 \quad \#$$

$$(N_0+1)^D = \sum_{i=0}^D \binom{D}{i} N_0^i$$

$$= 1 + D N_0 + \dots + D N_0^{D-1} + N_0^D$$

Δ This is larger than $(N_0^D + N_0^{D-1} + 1)$

So inequality still holds